Quantum Entanglement, Part 3

Lecture 7: Charged Particles, Field Transformations, and Relativistic Waves

Leonard Susskind

Transcript-derived notes curated by LazyingArt LLC
Produced with Video2Book

Chapter 1

Charged Particles, Field Transformations, and Relativistic Waves

Last time we began asking how a charged particle moves in electric and magnetic fields once the principles of relativity are imposed. The familiar Lorentz force law already contains the clue: its magnetic term depends on the particle's velocity, so different observers cannot agree separately about what is electric and what is magnetic. Our task is to formulate the law so that every inertial observer uses the same equation, even though the fields and the particle velocity change from frame to frame.

Unless stated otherwise, we use units in which $c = 1$ and the metric signature $(-,-,-,+)$. Spacetime components are ordered with the three spatial entries first:

$$\mu = (1, 2, 3, 0) = (x, y, z, t). \tag{1.1}$$

Latin indices m, n run only over the three spatial directions.

1.1 What survives of Newton's force law

In Newtonian mechanics,

$$\boldsymbol{F} = m\boldsymbol{a} \tag{1.2}$$

has the same form in every inertial frame because Galilean observers agree on acceleration. They also assign the same mass and force to the particle. Once the kinematics changes, there is no reason to expect this particular relation between force and ordinary acceleration to survive unchanged.

Electromagnetism and gravitation already warn us that relativistic extensions need not resemble one another. Coulomb's law and Newton's law of gravity are both inverse-square laws, but gravitation cannot be incorporated naturally into special relativity without a deeper change in spacetime geometry. Electromagnetism can. Indeed, magnetism is itself a relativistic effect: when factors of c are restored, a magnetic force carries a factor of velocity divided by c, and it disappears in the formal limit $c \to \infty$.

The durable part of Newton's law is the statement that force is the rate of change of momentum. For constant rest mass,

$$\boldsymbol{F} = m\frac{d\boldsymbol{v}}{dt} = \frac{d}{dt}(m\boldsymbol{v}) = \frac{d\boldsymbol{p}}{dt}. \tag{1.3}$$

Relativity will alter the relation between momentum and velocity, but it will preserve this way of organizing the equation.

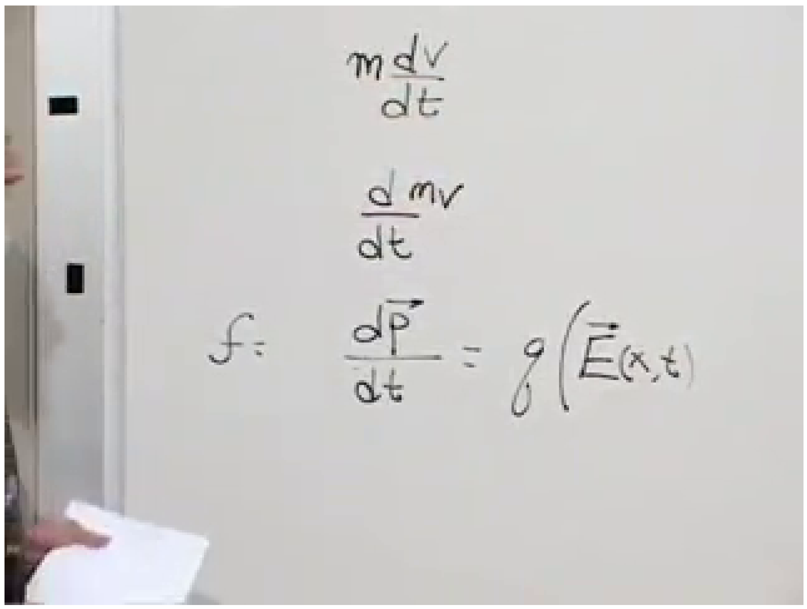

Figure 1.1: Force rewritten as $d\boldsymbol{p}/dt$ while the electric part of the Lorentz law is being placed on the board.

Lecture frame: 00:08:20.

In units $c = 1$, the pre-relativistic Lorentz force law is

$$\frac{d\boldsymbol{p}}{dt} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}). \tag{1.4}$$

Both fields may depend on position and time. The nonrelativistic limit is now the limit $v \ll 1$, not a limit in which the numerical constant $c = 1$ somehow changes. The electric term in (1.4) is already an ordinary vector. To understand the magnetic term in a form that can be carried into spacetime, we must look again at the cross product.

1.2 The cross product as an antisymmetric tensor

Take two three-dimensional vectors $\boldsymbol{A}$ and $\boldsymbol{B}$. Their outer product

$$T_{mn} = A_m B_n \tag{1.5}$$

is a rank-two tensor with nine components. It has no symmetry in general. Antisymmetrizing gives

$$K_{mn} = A_m B_n - A_n B_m, \qquad K_{mn} = -K_{nm}. \tag{1.6}$$

Its diagonal entries vanish, while every entry below the diagonal is fixed by the corresponding entry above it. A 3×3 antisymmetric tensor therefore has only three independent components.

That count is suggestive but not sufficient by itself. Three quantities do not automatically transform as a vector. In three dimensions, however, the three independent entries of (1.6) can be identified with the components of the cross product $\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$. The board first makes a temporary placement of C_2 in the upper-right position:

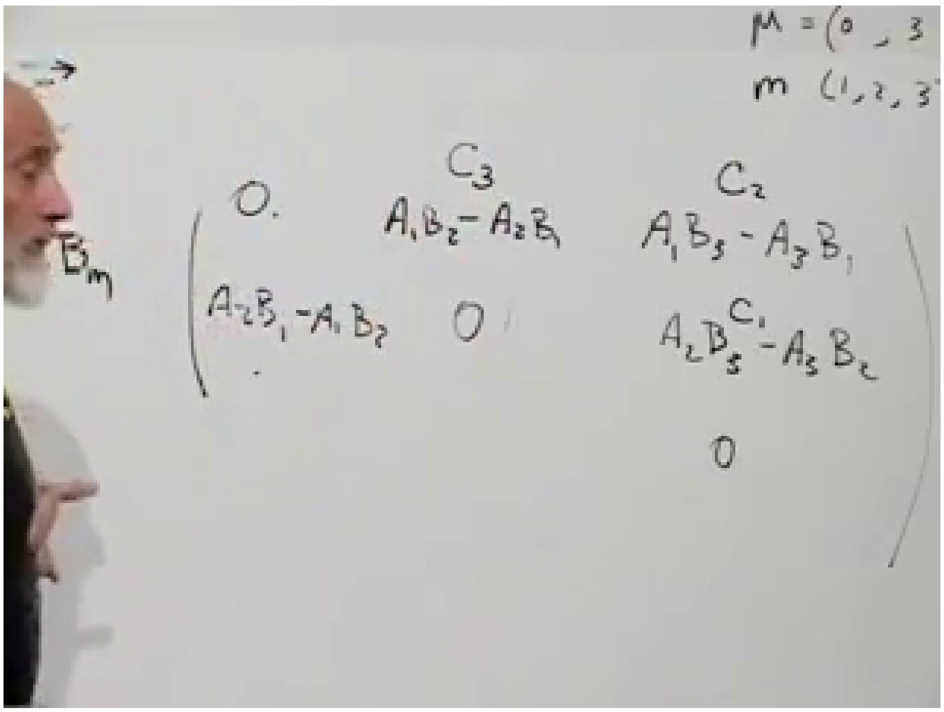

Figure 1.2: The antisymmetric product while its three independent entries are being matched to the components of $\boldsymbol{C}$. The upper-right label is still provisional here.

Lecture frame: 00:17:33.

Cycling $1 \to 2 \to 3 \to 1$ gives the component formulas

$$C_3 = A_1B_2 - A_2B_1, \tag{1.7}$$
$$C_1 = A_2B_3 - A_3B_2, \tag{1.8}$$
$$C_2 = A_3B_1 - A_1B_3. \tag{1.9}$$

The last line shows that the upper-right entry is $-C_2$, not C_2. The correction is part of the argument rather than a detail to hide.

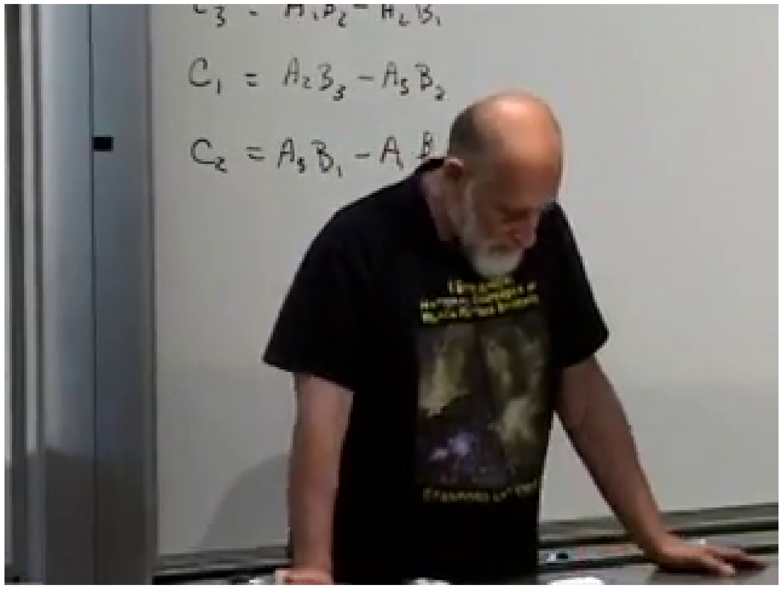

Figure 1.3: The completed cyclic component identities that fix the temporary sign in the antisymmetric array.

Lecture frame: 00:21:54.

The resulting matrix is

$$K_{mn} = \begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix}. \tag{1.10}$$

There is a useful geometry behind this notation. A component with indices $1, 2$ belongs naturally to the xy-plane and is identified with a vector component along the perpendicular z-axis. One may describe the same oriented object either by a plane or by an arrow normal to that plane. The arrow description is special to three dimensions; in higher dimensions an antisymmetric rank-two tensor still exists, but it no longer has the same number of components as a vector. Thus the tensor, not the perpendicular arrow, is the form of the cross product that generalizes.

Equivalently, after choosing an orientation,

$$C_k = \frac{1}{2}\epsilon_{kmn}K_{mn}. \tag{1.11}$$

The need to choose an orientation is the source of the right-hand rule and of the familiar distinction between polar and axial vectors. [1]

[1]Editorial clarification: The Levi–Civita formula and the axial-vector terminology are standard notation for the plane-versus-normal geometry developed explicitly in the lecture.

1.3 The magnetic field in tensor form

The same dual description applies to the magnetic field. We define its antisymmetric spatial tensor by

$$B_{mn} = \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix}. \tag{1.12}$$

The extra overall sign relative to (1.10) is a convention. Once it has been chosen, it must be used consistently; changing the sign of all three vector components would describe the same tensor-to-vector identification with the opposite orientation convention.

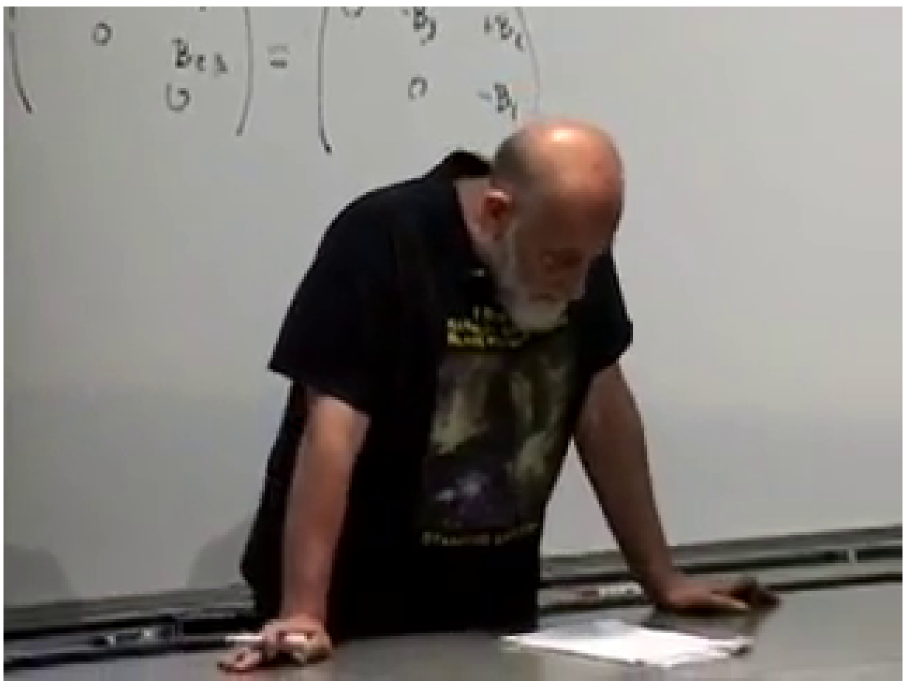

Figure 1.4: The magnetic field written both as an antisymmetric spatial array and in terms of its three vector components.

Lecture frame: 00:24:40.

Now take the third component of the cross product:

$$(\boldsymbol{v} \times \boldsymbol{B})_3 = v_1 B_2 - v_2 B_1 \tag{1.13}$$

$$= v_1 B_{13} + v_2 B_{23} + v_3 B_{33}. \tag{1.14}$$

The last term may be added because $B_{33} = 0$. Cycling the free component gives the compact rule

$$(\boldsymbol{v} \times \boldsymbol{B})_m = v_n B_{nm}, \tag{1.15}$$

where the repeated index n is summed. The magnetic force is therefore

$$F_m^{(B)} = q v_n B_{nm}. \tag{1.16}$$

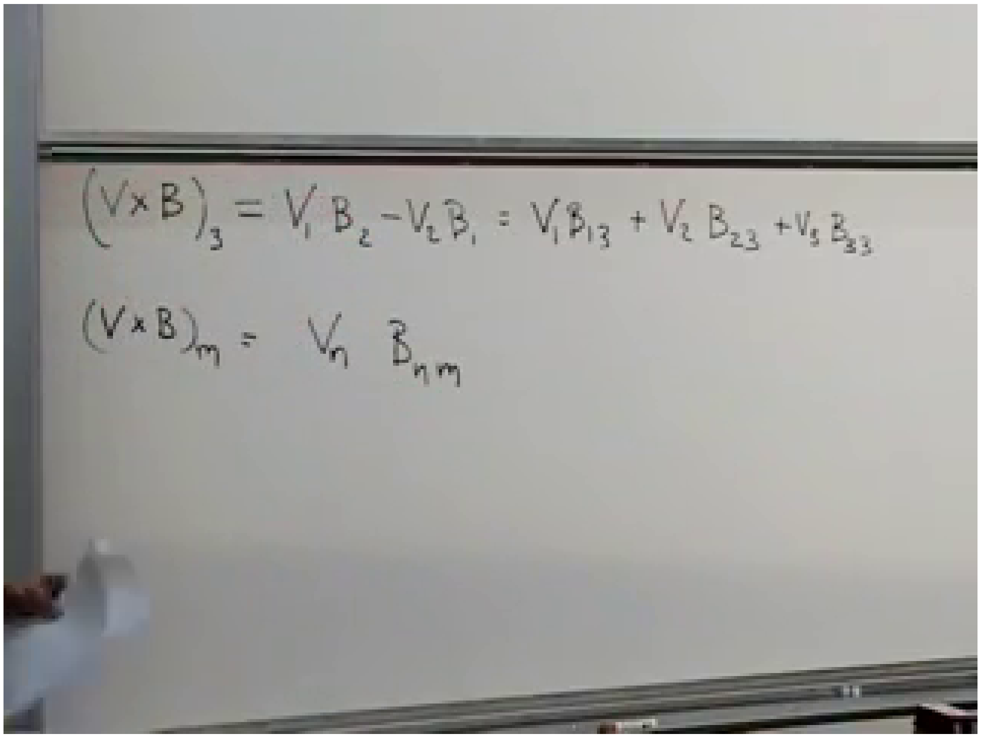

Figure 1.5: The third component of $\boldsymbol{v} \times \boldsymbol{B}$ and its general expression as a contraction with B_{nm}.
Lecture frame: 00:28:30.

Question from the audience. Does the free index m label a separate magnetic component of the force?

Response. No. It labels whichever spatial component of the force is being calculated. The electric field has simply been set to zero for the present argument, and every index here runs only over the three spatial directions.

The tensor notation makes the no-work property of a magnetic field nearly automatic. Differentiate the squared speed:

$$\frac{d}{dt}(v_m v_m) = 2\dot{v}_m v_m \tag{1.17}$$
$$\propto 2(\boldsymbol{v} \times \boldsymbol{B})_m v_m \tag{1.18}$$
$$= 2v_n B_{nm} v_m. \tag{1.19}$$

Terms with $m = n$ vanish because the diagonal of B_{nm} is zero. Every off-diagonal term has a partner with m and n interchanged; the two cancel because $v_n v_m$ is symmetric while B_{nm} is antisymmetric. Hence

$$\frac{d}{dt}(v_m v_m) = 0. \tag{1.20}$$

A pure magnetic force bends a trajectory without changing its speed or kinetic energy.

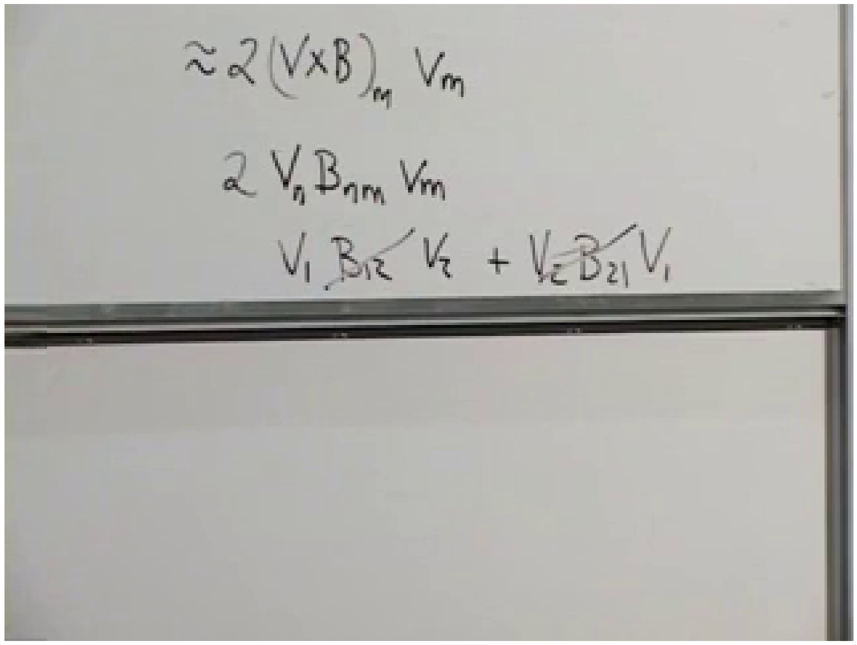

Figure 1.6: The paired terms in $v_n B_{nm} v_m$ cancel because the magnetic tensor changes sign when its indices are interchanged.

Lecture frame: 00:35:40.

This proof is intentionally more elaborate than saying that a cross product is perpendicular to its first factor. The latter statement is peculiar to the three-dimensional vector representation. The contraction of a symmetric pair with an antisymmetric tensor is the argument that will survive in four dimensions.

Question from the audience. Must the magnetic field be time independent for the speed to remain constant?

Response. The algebra above never differentiates the field and does not require $\partial_t \boldsymbol{B} = 0$. A magnetic term by itself still does no work. In electrodynamics, however, a changing magnetic field produces an electric field, and that electric field can change the particle's speed.

1.4 Why electric and magnetic fields must mix

The mixing of $\boldsymbol{E}$ and $\boldsymbol{B}$ can be seen before any four-dimensional calculation. Consider slow motion, so that ordinary Newtonian statements about acceleration are adequate. Let a magnetic field point along the y-axis and let a charged particle move along the x-axis. The magnetic force points along the z-axis:

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}. \tag{1.21}$$

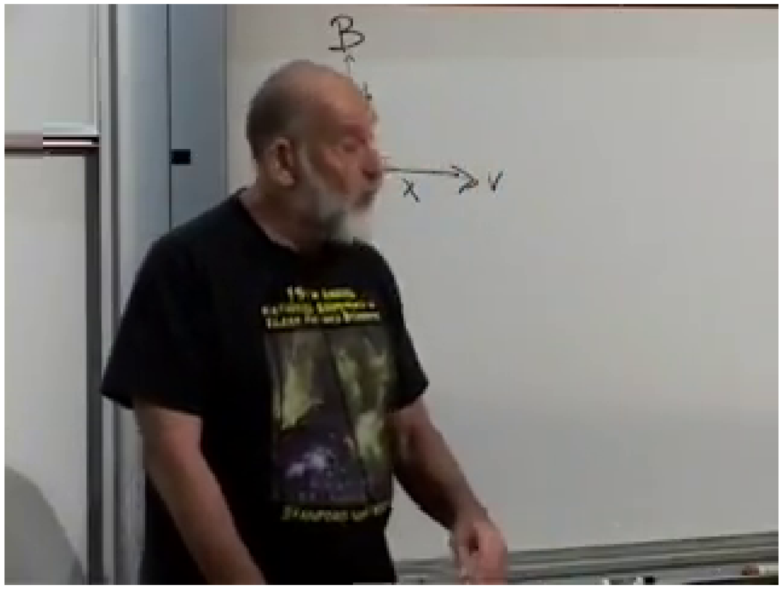

Figure 1.7: The board sketch for a particle moving across a magnetic field; the three directions are used to compare the laboratory and co-moving descriptions.

Lecture frame: 00:42:20.

The geometry may be redrawn without the perspective of the classroom:

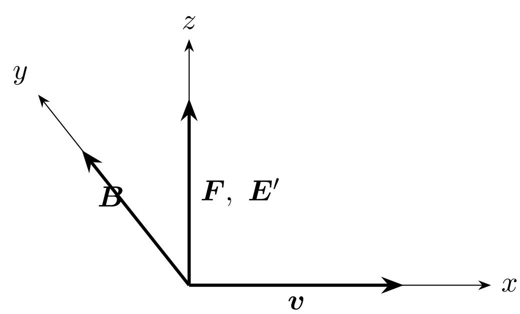

Figure 1.8: In the laboratory the z-directed force is magnetic; in the instantaneous co-moving frame the same force must be electric.

Now introduce an observer moving with the particle at that instant. The particle's velocity in the new frame is zero, so the magnetic term $q\boldsymbol{v}' \times \boldsymbol{B}'$ vanishes. Yet in the slow-motion limit both observers must agree that the particle has a z-acceleration. If the Lorentz force law has the same form in both frames, the co-moving observer must attribute the force to an electric field $\boldsymbol{E}'$ along z. A field that was purely magnetic in the laboratory is therefore not purely magnetic in the moving frame.

This is the physical reason to seek one object whose components include both fields. A change of frame does not merely rotate $\boldsymbol{E}$ among its own three components and $\boldsymbol{B}$ among its own three components; it mixes the two sets.

1.5 The electromagnetic field tensor

Together the electric and magnetic fields have six components. A four-dimensional antisymmetric tensor also has six independent components, since

$$\frac{4(4-1)}{2} = 6. \tag{1.22}$$

This does not mean that an antisymmetric tensor is literally the only collection of six quantities one could invent: six scalars, or a vector and two scalars, would also contain six numbers. It is the natural geometrical object whose components can mix in the required way.

To keep the board convention explicit, we display the contravariant tensor in the spatial-first order $(1, 2, 3, 0)$:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}. \tag{1.23}$$

The upper-left 3×3 block is precisely the magnetic tensor (1.12); components with one spatial and one time index contain the electric field. Antisymmetry supplies the remaining entries.

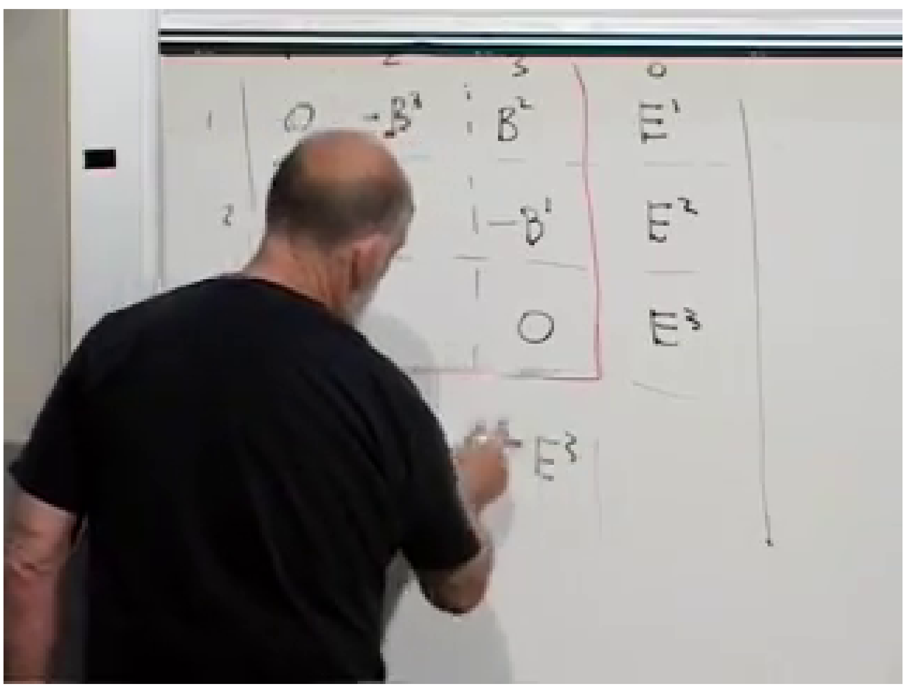

Figure 1.9: The magnetic spatial block and electric time-space column being assembled into the four-dimensional field tensor.

Lecture frame: 00:46:35.

Question from the audience. When should these field components carry upper rather than lower indices?

Response. For the three spatial labels alone, the board sometimes suppresses the distinction while identifying the familiar Maxwell–Faraday fields. In a covariant calculation the distinction must be restored: raising or lowering a spatial index changes a sign in the chosen metric convention.

What is invariant is not a particular arrangement of signs on the board, but the antisymmetric spacetime tensor together with a consistent metric and index convention. The notation F may suggest Faraday, although the lecture does not claim that historical attribution with confidence. Einstein worked out the transformation of electric and magnetic components in 1905; the compact four-dimensional tensor language came with the later spacetime formulation associated with Minkowski.

1.6 Four-velocity and four-force

The particle side of the equation must also be made covariant. Proper time τ is invariant, so differentiating the position four-vector with respect to it gives another four-vector:

$$u^\mu = \frac{dx^\mu}{d\tau} = \gamma(v_x, v_y, v_z, 1), \qquad \gamma = \frac{1}{\sqrt{1-v^2}}. \tag{1.24}$$

The last component follows from $dt/d\tau = \gamma$. The four-momentum is

$$p^\mu = mu^\mu, \tag{1.25}$$

whose spatial part is the relativistic momentum $\boldsymbol{p} = m\gamma\boldsymbol{v}$.

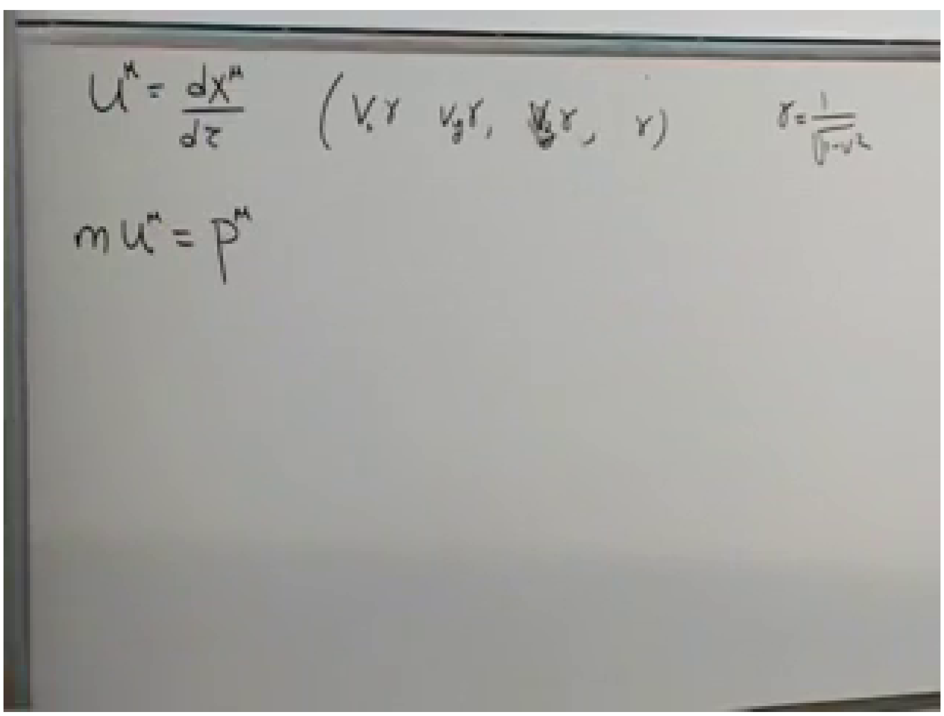

Figure 1.10: Four-velocity, its components, the Lorentz factor, and four-momentum.

Lecture frame: 00:50:30.

Because p^μ is a four-vector and τ a scalar, the natural relativistic left-hand side is

$$f^\mu \equiv \frac{dp^\mu}{d\tau}. \tag{1.26}$$

The four-force cannot be chosen arbitrarily. Four-velocity has fixed norm,

$$u_\mu u^\mu = 1. \tag{1.27}$$

Differentiating along the world line gives

$$0 = \frac{d}{d\tau}(u_\mu u^\mu) \tag{1.28}$$
$$= \frac{du_\mu}{d\tau}u^\mu + u_\mu \frac{du^\mu}{d\tau} \tag{1.29}$$
$$= 2u_\mu \frac{du^\mu}{d\tau}. \tag{1.30}$$

For constant mass this implies

$$f^\mu u_\mu = 0. \tag{1.31}$$

Thus four-force is Minkowski-orthogonal to four-velocity. This is a purely kinematic constraint following from the normalization of u^μ, not a special property of electromagnetism.

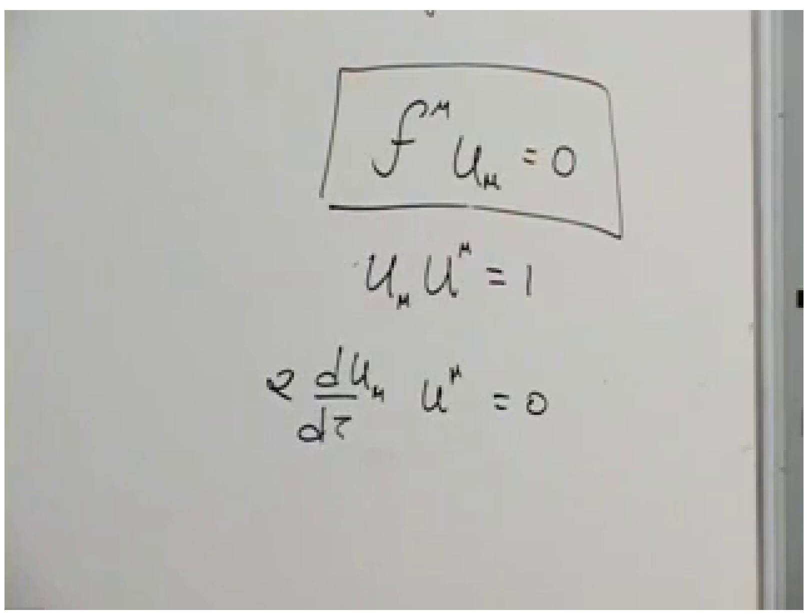

Figure 1.11: The unit norm of four-velocity and the resulting orthogonality of four-force and four-velocity.

Lecture frame: 00:56:50.

The three-dimensional magnetic calculation has already supplied the trick for satisfying (1.31). Contract an antisymmetric tensor with one copy of the velocity:

$$\boxed{\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu.} \tag{1.32}$$

Indeed,

$$f^{\mu} u_{\mu} = qF^{\mu\nu} u_{\nu} u_{\mu} = 0, \tag{1.33}$$

because $F^{\mu\nu}$ is antisymmetric and $u_{\nu} u_{\mu}$ is symmetric. This is not the only conceivable relativistic force law, but it is the simplest one linear in both the electromagnetic field and the velocity.

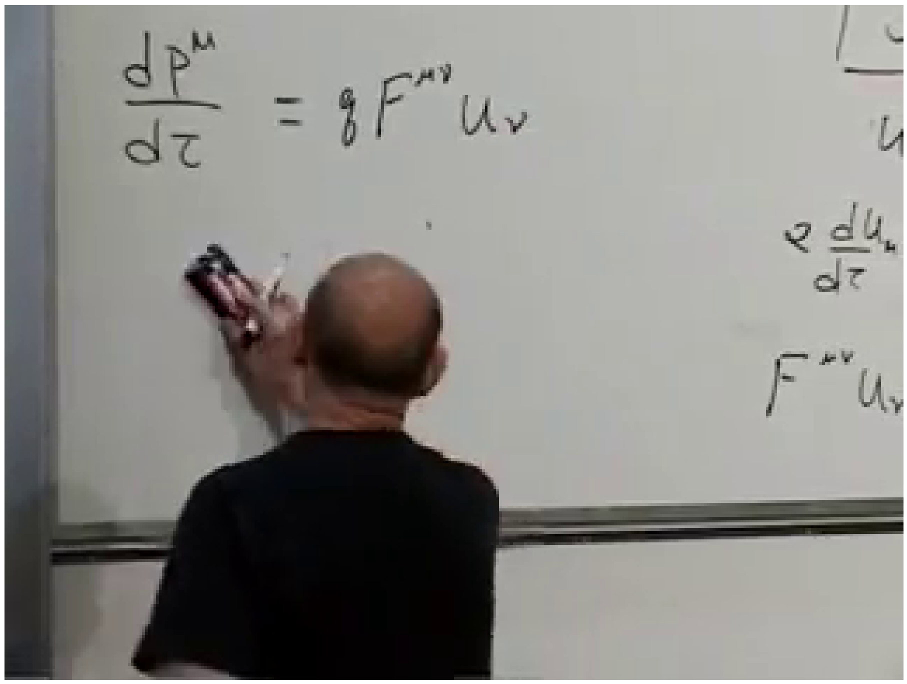

Figure 1.12: The covariant Lorentz-force equation on the left and its vanishing contraction with a second four-velocity on the right.

Lecture frame: 01:00:00.

1.7 Recovering the ordinary Lorentz force

A covariant-looking equation must still reproduce the known force law. With the metric convention stated at the beginning,

$$u_{\mu} = \gamma(-v_x, -v_y, -v_z, 1). \tag{1.34}$$

For the x-component, (1.23) and (1.32) give

$$\frac{dp^1}{d\tau} = q(F^{12} u_2 + F^{13} u_3 + F^{10} u_0) \tag{1.35}$$

$$= q\gamma(B_3 v_y - B_2 v_z + E_1) \tag{1.36}$$

$$= q\gamma[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}]_x. \tag{1.37}$$

The other spatial components follow by cycling the indices.

Question from the audience. Where is the qE_x term in the component expansion?

Response. It is the term $qF^{10} u_0$. A field-tensor component with one spatial and one time index is an electric field component, and $u_0 = \gamma$.

The factor γ on the right is not an extra force. Since

$$\frac{dp^i}{d\tau} = \frac{dt}{d\tau}\frac{dp^i}{dt} = \gamma\frac{dp^i}{dt}, \tag{1.38}$$

it cancels from both sides of (1.37). The exact spatial equation is therefore again

$$\frac{d\boldsymbol{p}}{dt} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}), \qquad \boldsymbol{p} = m\gamma\boldsymbol{v}. \tag{1.39}$$

The familiar form survives, but the relativity is hidden in the momentum.

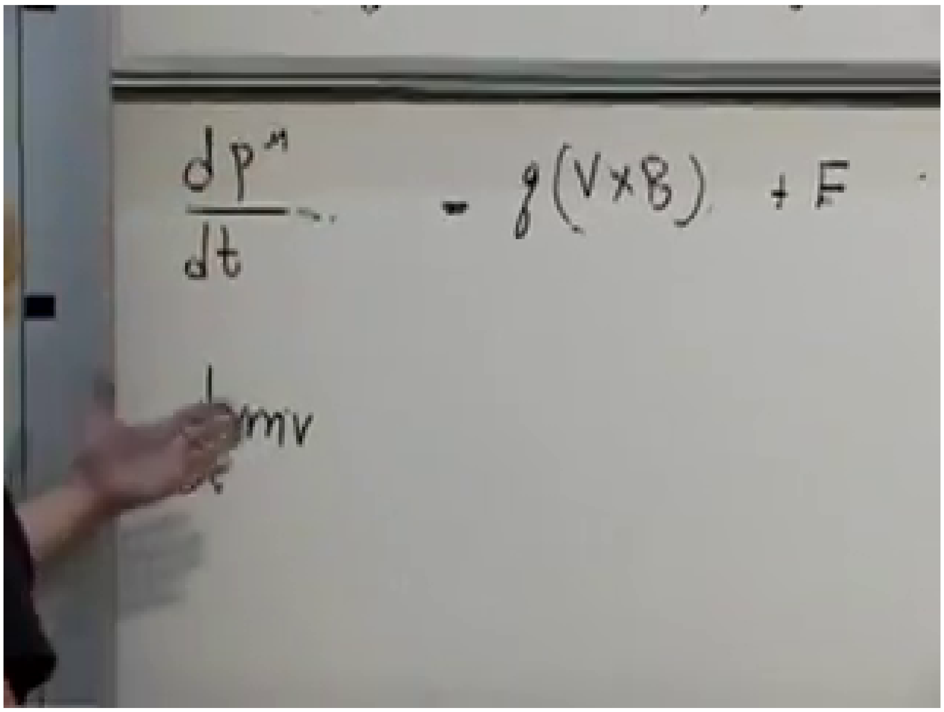

Figure 1.13: The three-dimensional Lorentz law recovered after the common proper-time factor is removed; the momentum retains its factor of γ.

Lecture frame: 01:06:30.

The time component carries the corresponding energy statement,

$$\frac{dp^0}{dt} = q\boldsymbol{E} \cdot \boldsymbol{v}, \tag{1.40}$$

so a magnetic field performs no work while an electric field may change the particle energy. [2]

Question from the audience. Is this derivation mathematics or physics?

Response. It is both. Tensor algebra supplies the compact form, but substantial physics entered through the principle that all inertial frames obey the same laws and through the assumption that the equations of motion are second order. A second-order law predicts the future from position and velocity; a third-order law would require acceleration as additional initial data. Experiment, not notation, tells us which kind of law nature uses.

[2] Editorial clarification: The time-component check is the direct companion of the lecture's spatial calculation; it is included to expose the same no-work property in four-dimensional notation.

1.8 How the fields transform

Equation (1.32) has the same form in every frame because both sides are four-vectors. The field tensor itself transforms with one Lorentz matrix for each index:

$$F'^{\mu\nu} = \Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}F^{\rho\sigma}. \tag{1.41}$$

The board derives this rule by first transforming a simple product $A^\mu B^\nu$. Once the rule is known for a product of vectors, linearity gives the rule for any rank-two tensor.

For a boost along the 1-axis, the coordinates transform as

$$x'^1 = \gamma(x^1 - vx^0), \qquad x'^0 = \gamma(x^0 - vx^1), \tag{1.42}$$

$$x'^2 = x^2, \qquad x'^3 = x^3. \tag{1.43}$$

Apply the same transformation separately to both tensor indices. For the component parallel to the boost, the board begins with a product:

$$(A'^0 B'^1) = \gamma^2(A^0B^1 - vA^0B^0 - vA^1B^1 + v^2A^1B^0). \tag{1.44}$$

Replacing each product by the corresponding tensor component and then using antisymmetry, $F^{00} = F^{11} = 0$ and $F^{10} = -F^{01}$, gives

$$F'^{01} = \gamma^2(1 - v^2)F^{01} = F^{01}. \tag{1.45}$$

Thus the electric field component parallel to the motion is unchanged. The parallel magnetic component behaves similarly.

The transverse components are different. Following the primed-frame and sign convention used on the board,

$$F'^{02} = \gamma(F^{02} - vF^{12}), \tag{1.46}$$

$$E'_2 = \gamma(E_2 + vB_3). \tag{1.47}$$

A transverse electric component acquires part of a magnetic component. The remaining transformation laws follow from the same two-index rule and are left as the lecture's exercise.

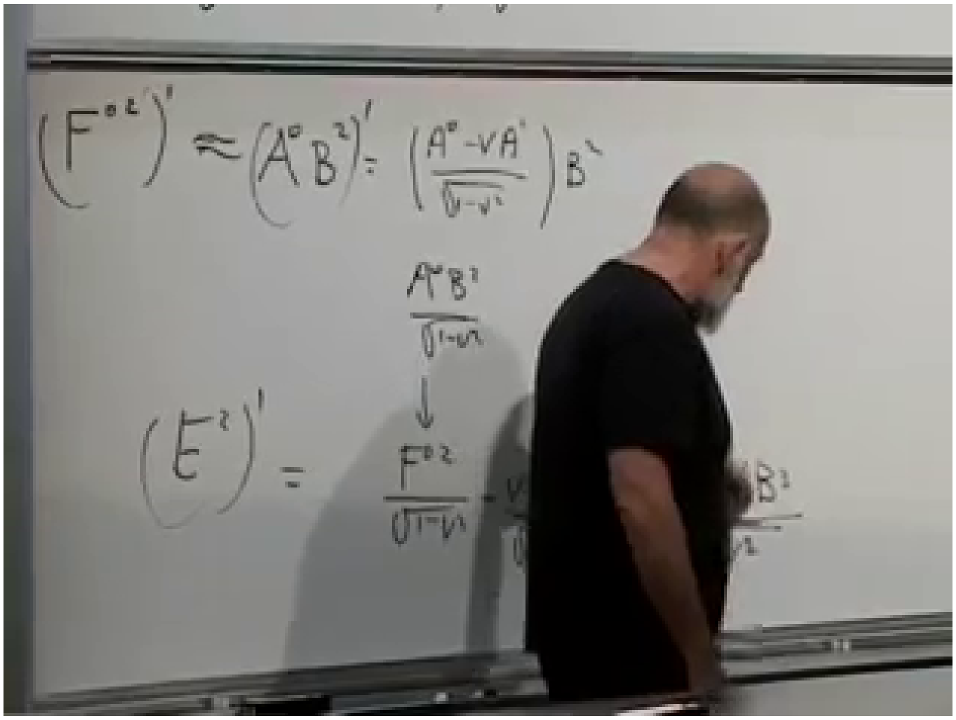

Figure 1.14: The transverse tensor transformation at the point where the E_2' component is seen to contain a term proportional to vB_3.

Lecture frame: 01:23:40.

The sign in (1.47) depends on which frame is declared to move with $+v$, as well as on index and metric conventions. Its invariant content does not: transverse electric and magnetic components mix, and the covariant force equation remains valid after the transformation.

Question from the audience. For a current in a wire, can one move at some velocity that makes the magnetic field vanish?

Response. Moving with the conduction electrons is not enough because the positive lattice charges then move in the new frame. A neutral current-carrying wire is therefore subtler than a single moving lump of charge. A purely magnetic field cannot be transformed completely away. If electric and magnetic fields are both present, their transformed pieces can cancel in special circumstances; for one uniformly moving charge, the charge's rest frame is the simple frame in which its magnetic field vanishes.

The precise invariant test uses

$$\boldsymbol{B}^2 - \boldsymbol{E}^2, \qquad \boldsymbol{E} \cdot \boldsymbol{B}. \tag{1.48}$$

A frame with $\boldsymbol{B}' = 0$ can exist only for an electric-dominated field with $\boldsymbol{E} \cdot \boldsymbol{B} = 0$; a field that is purely magnetic in one frame is magnetic-dominated and cannot meet that condition. [3]

Question from the audience. Do the electric scalar potential and magnetic vector potential form a four-vector?

Response. Yes. With convention-dependent signs they combine into the four-potential A^μ. Maxwell's equations will make systematic use of that object.

[3]Editorial clarification: These invariant criteria state precisely the qualification reached in the classroom discussion; their derivation is deferred.

1.9 Derivatives of fields

We now leave the trajectory of one particle and consider a field throughout spacetime. Let

$$\phi = \phi(x^1, x^2, x^3, x^0) \tag{1.49}$$

be a scalar field. Its four derivatives form the covariant components of a four-vector:

$$\partial_\mu \phi \equiv \frac{\partial \phi}{\partial x^\mu} = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z}, \frac{\partial \phi}{\partial t} \right). \tag{1.50}$$

The chain rule applied to the Lorentz coordinate transformation proves this covariant transformation law. Raising the index reverses the spatial signs but leaves the time sign unchanged:

$$\partial^\mu \phi = \left(-\frac{\partial \phi}{\partial x}, -\frac{\partial \phi}{\partial y}, -\frac{\partial \phi}{\partial z}, \frac{\partial \phi}{\partial t} \right). \tag{1.51}$$

Every differentiation contributes a lower index. Thus $\partial_\nu V^\mu$ is a rank-two tensor with one upper and one lower index, and differentiating a tensor produces an object with one additional lower index. This is the basic rule needed to write field equations without choosing a preferred inertial frame.

1.10 The invariant scalar wave equation

The simplest wave equations contain two derivatives. If the equation is to have the same form in every inertial frame, its left-hand side should be a scalar. Contracting the derivative indices gives the unique simplest choice:

$$\boxed{\partial^\mu \partial_\mu \phi = 0.} \tag{1.52}$$

Writing out the components in the present signature,

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} - \frac{\partial^2 \phi}{\partial z^2} = 0. \tag{1.53}$$

The contraction in (1.52) makes Lorentz invariance manifest; equation (1.53) is the same statement written component by component.

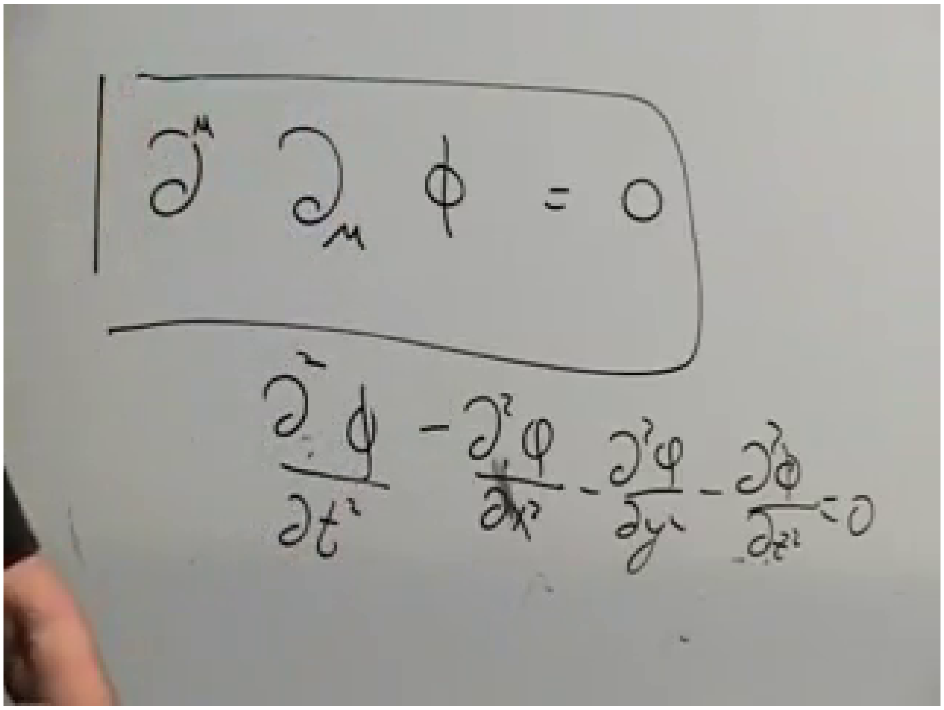

Figure 1.15: The contracted derivative form of the scalar wave equation and its time-minus-space component expansion.

Lecture frame: 01:39:40.

Question from the audience. Must the right-hand side of the wave equation be zero?

Response. No. It may be any scalar compatible with the problem. A constant times ϕ, or a nonlinear scalar function such as one proportional to ϕ^2, still preserves covariance. Zero gives the simplest free wave equation.

The point of this scalar example is preparation. Maxwell's equations mix electric and magnetic fields more intricately, but they too can be written so that Lorentz invariance is visible from their index structure. That is the problem to which we return next time.